ASRS–HFACS Project — Team Understanding & Viva Prep Guide

> Purpose: every team member should be able to answer "what are we doing, why, which paper did we use for what, and what does each file do" without opening the code. Items marked **[CONFIRM]** are things only your team knows (final numbers, exact class names, who did what). Fill them in before the evaluation.

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1. The 30-Second Pitch (everyone memorizes this)

> "Aviation safety reports are written as free text by pilots and investigators. We built a system that reads such a report and automatically classifies **which human-factor failure** was involved (for example a perceptual error, a decision error, a skill-based error or a violation), using the **HFACS** framework. We trained on **ASRS** (voluntary pilot reports) and tested whether it still works on **NTSB** (official accident investigation reports) — that is our **cross-domain transfer** contribution. We compare classical ML, transformers, and LLM reasoning, and we make the model **explainable** so a safety analyst can see *why* a report got a label."

One-line version: *Explainable NLP that tags human-error causes in aviation reports, and checks whether a model trained on one kind of report works on another.*

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2. Key Terms (basic questions teachers love)

Term	Simple meaning
ASRS	Aviation Safety Reporting System (NASA). Pilots/controllers *voluntarily* report incidents and near-misses as narratives. Large, rich text, but self-reported.
NTSB	National Transportation Safety Board (USA). Official investigation reports of accidents. Externally investigated, so closer to "ground truth".
HFACS	Human Factors Analysis and Classification System. A standard taxonomy for categorizing *why* humans contributed to an accident (errors, violations, preconditions, supervision, organization).
Weak labeling	Creating training labels automatically with rules/clustering instead of paying humans to label everything. Labels are noisy but cheap.
Cross-domain transfer	Train on domain A (ASRS), test on domain B (NTSB), to see if the model generalizes.
TF-IDF	Classical text features: words weighted by how distinctive they are for a document.
Word2Vec / Doc2Vec	Learned word/document embeddings (dense vectors capturing meaning).
DistilBERT / SafeAeroBERT	Transformer models. SafeAeroBERT is BERT further pre-trained on aviation safety text (ASRS + NTSB).
LLM-CoT (HFACS-CoT+)	Using an LLM with a step-by-step question chain that mirrors how a human investigator decides the HFACS class.
Micro-F1 / Macro-F1	Micro-F1 = overall performance weighted by sample count. Macro-F1 = average over classes equally (reveals weakness on rare classes).
SHAP	Method that shows how much each word pushed the model toward a label (explainability).
HDBSCAN / UMAP	UMAP shrinks embeddings to fewer dimensions; HDBSCAN finds clusters without needing a preset cluster count.

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3. Problem Statement & Motivation

Problem. Thousands of aviation safety narratives are produced every year. Reading them to find the *human-factor* cause is slow, subjective, and needs trained experts. Different experts can label the same report differently.

Why it matters. Most accidents have a human-factor component. If we can tag causes at scale, safety teams can spot recurring failure patterns (e.g., repeated perceptual errors during approach) and act on them before the next accident.

The gap we address (what makes it a "thing to work upon"):

1. **Labels are scarce.** ASRS narratives are abundant but not labeled in HFACS. Expert labeling is expensive. → we use *weak labeling*.
2. **Domain shift.** Models trained on voluntary reports may fail on official investigation reports because the two are written and reported differently. This is rarely tested directly. → we run an explicit **ASRS → NTSB transfer test**.
3. **Trust.** Safety is a high-stakes field; a black-box label is not acceptable. → we add **explainability**.

4. **Which method actually works?** Papers use different models on different data. → we compare classical ML, transformers, and LLM reasoning in **one unified table** on the same data.

How we thought of it (honest storyline you can tell the teacher): We started from the idea of applying NLP to safety text, found that HFACS is the standard human-factor taxonomy, then found that existing papers each solve only a piece (weak labels, embeddings, domain-specific BERT, LLM reasoning, phase analysis). Nobody combined them with a cross-domain test, so we organized a build plan that takes the best technique from each paper and puts them in one pipeline. **[CONFIRM: adjust to how your team genuinely arrived at the topic.]**

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4. System Overview (the pipeline in one picture)

Stages used in the plan: Stage 0 = data & labels, Stage 1 = baseline, Stage 2 = transformers & extra baselines, Stage 3–4 = explainability, tuning, flight-phase analysis, writeup.

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5. Research Papers — What We Took From Each

Paper (short name)	What it is	What we borrowed	Used in file(s)
Liu, Li, Ng, Han & Feng (2025) — HFACS-guided Chain-of-Thoughts	LLM reasoning for accident investigation; builds the **GAHFACS** dataset from NTSB	How the NTSB dataset was constructed (filter far-part 091, match narratives and findings via `ev-id`, human experts recode into HFACS 8.0); the 4-question decision chain; the **HFACS-CoT+** idea (one LLM call per question)	`data_loading.py`, `llm_cot_baseline.py`, final report
Madeira, Melício, Valério & Santos (2021) — ML & NLP for prediction of human factors in incident reports	Classical NLP pipeline + weak labeling for human factors	The **consistency criterion** for weak labels (12 of 15 agree, else discard keyword); keyword→label seed table; 3-stage preprocessing; TF-IDF / Word2Vec / Doc2Vec; Micro-F1 as primary metric; random search + fANOVA	`keyword_weak_labeler.py`, `preprocessing.py`, `baseline_models.py`, `evaluation.py`, `hyperparameter_tuning.py`
Macedo, Ramos, Maior, Moura & Lins (2026) — BERT classifier + topic modeling for pilot failures	Unsupervised discovery of failure themes	Pipeline **BERT mean-pooled embeddings → UMAP → HDBSCAN → class-based TF-IDF** to name clusters; hyperparameter sensitivity (min_cluster_size, min_samples vs. outlier rate)	`weak_labeler_cluster.py`
Shi, Zurada, Cao & Guan (2022) — classical vs. deep text classification on ASRS	Compares feature extractors on ASRS	Finding that TF-IDF slightly beats Word2Vec for classical models, but Word2Vec + deep models do better; preprocessing details	`preprocessing.py`, `baseline_models.py`
Vempati, Woods & Solano (2023) — human error analysis of ASRS	Descriptive statistics of error types in ASRS	Published ASRS error distribution (Perceptual 53%, Decision 32%, Skill-Based 26%; CRM precondition 53%) used as a **sanity check**; weighted keyword table for flight phase; cross-tab figure style; argument for why ASRS→NTSB transfer may degrade	`evaluation.py`, `flight_phase_analysis.py`, discussion
Andrade & Walsh (2023) — SafeAeroBERT	BERT further pre-trained on ASRS + NTSB	Domain-adapted checkpoint, binary-relevance fine-tuning (one output per label with binary cross-entropy), per-class reporting, SHAP context	`transformer_model.py`, `explainability.py`

Teacher-proof one-liner for each paper:

- *Liu:* "Gave us NTSB ground truth and the LLM reasoning chain."
- *Madeira:* "Gave us the weak-labeling rule, preprocessing order, baselines and tuning method."
- *Macedo:* "Gave us the clustering-based way to discover labels without supervision."
- *Shi:* "Told us what to expect from TF-IDF vs Word2Vec on ASRS."
- *Vempati:* "Gave us the reference ASRS statistics and the explanation of why transfer may fail."
- *Andrade & Walsh:* "Gave us the aviation-specific BERT and the explainability angle."

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6. File-by-File Guide (which file represents what)

Week 1 — Data foundation & labeling (Stage 0)

`data_loading.py` — *Gets all data into one shape.*

- `load_ntsb()` reads NTSB raw tables (`ev-id`, `narr_accp`, `narr_accf`, findings) following the structure used by Liu et al. (filtered to far-part 091).
- `standardize_schema()` converts ASRS or NTSB into the common schema: `report_id, source, text, label`.
- *Why:* ASRS and NTSB look different; the transfer test only works if both share one schema.
- *If asked "why not guess column names?"* → we followed how a published dataset was actually built, so our NTSB handling is citable and reproducible.

`keyword_weak_labeler.py` — *Automatic labels from keywords, with a quality gate.*

- For a candidate keyword, sample 15 documents containing it; if at least 12 agree on one HFACS class, label **all** documents with that keyword; otherwise discard the keyword.
- Logs every keyword, pass/fail, and number of documents labeled (becomes a table in the report).
- *Why:* We have no large hand-labeled ASRS set. The gate prevents forcing noisy labels.

- *Limitation to admit:* weak labels are not human ground truth; they are an approximation.

`weak_labeler_cluster.py` *(optional second labeler) — Let the data suggest the groups.*

- BERT embeddings → UMAP → HDBSCAN → class-based TF-IDF names each cluster (top 5 words).
- Small grid search over `min_cluster_size` and `min_samples`, tracking % of unassigned outliers.
- Clusters are manually mapped to HFACS classes and saved to a JSON config.
- *Why:* A second, independent way to label; agreement between the two labelers would raise confidence. **[CONFIRM: whether you implemented/used it.]**

Week 2 — Baseline pipeline & cross-domain test (Stage 1)

`preprocessing.py` — *Text cleaning in a fixed, documented order.*

- Clean (remove duplicates, drop non-representative incidents) → normalize (lowercase, strip punctuation, translate if needed) → tokenize alphabetic-only → remove standard + domain stopwords (summary, probable, cause, accident, contribute, factor, find, conclusion…) → lemmatize → drop words appearing ⩽ 5 times.
- *Why domain stopwords?* Words like "probable cause" appear in every report and carry no signal for class.

`baseline_models.py` — *The simple, strong reference models.*

- Feature extractor switch: `FEATURE_METHOD = "tfidf" | "word2vec" | "doc2vec"`.
- Classifiers: Logistic Regression and LinearSVC.
- *Why:* A baseline tells us whether the fancy models are actually worth it.

`evaluation.py` — *How we measure.*

- Micro-F1 (primary), Macro-F1 and Precision (secondary). Accuracy alone is misleading with class imbalance.
- **Cross-domain sanity check:** prints ASRS vs NTSB per-class distribution, compares with Vempati's published ASRS numbers, logs it right before the ASRS → NTSB transfer test.

Week 3 — Transformers & extra baselines (Stage 2)

`transformer_model.py` — *Deep-learning classifier.*

- `BASE_CHECKPOINT` switch: DistilBERT vs SafeAeroBERT, same fine-tuning loop.
- Binary relevance: one output per label with binary cross-entropy (a report can have more than one human factor), instead of one softmax head.
- Output per-class precision/recall/F1 to see *where* domain pre-training helps (the SafeAeroBERT paper found it helped weather/procedure but not human-factors/aircraft).

`llm_cot_baseline.py` — *Reasoning-style classifier using an LLM.*

- Four questions: Q1 did the individual do/fail to do something? → Q2 performance/skill error? → Q3 judgment/decision error? → Q4 known deviation (asked only if Q2 and Q3 are both "no").
- **HFACS-CoT+:** each question is a separate LLM call (the paper's ablation showed this fixes logical inconsistencies and improves F1/recall).
- *Why it matters:* needs no training labels, and its reasoning is readable.

`model_comparison.py` — *The master results table.*

- One function that every model reports into: `model | domain (in/cross) | Micro-F1 | Macro-F1 | Precision | Recall`.
- This is the centerpiece of the Results section.

Week 4 — Explainability, tuning, phase analysis, writeup (Stage 3–4)

`explainability.py` — *Why did the model say that?*

- SHAP on whichever transformer is active; top-weighted words per class for Logistic Regression as the simpler comparison.

`hyperparameter_tuning.py` — *Systematic tuning instead of guessing.*

- Random search (adapted ranges), then an importance ranking (fANOVA or a variance proxy) after ~30–50 trials. Madeira et al. found only 2–3 of 7 hyperparameters really mattered.
- Bayesian optimization stopping rule: gains plateaued past ~4 free variables.

`flight_phase_analysis.py` *(bonus) — Where in the flight do errors happen?*

- Seed keywords from Vempati (runway, landing, approach, pattern, tower, winds…), extended with taxi, takeoff, climb, cruise, descent, final, go-around, touchdown.
- Heatmap of error type × flight phase.

Final report / cross-domain discussion — Vempati's argument: voluntary reporters tend to over-report perceptual errors and under-report their own decision/skill failures, unlike externally investigated NTSB cases. Liu et al.'s LLM-vs-human comparison is the template for comparing our LLM-CoT baseline with human judgment.

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7. Our Contribution (what we claim, and what we don't)

We claim:

1. A single reproducible pipeline combining weak labeling, classical baselines, transformers and LLM reasoning on aviation human-factor text.
2. An explicit **ASRS → NTSB transfer evaluation** with a distribution sanity check.
3. A **unified comparison table** across model families, reported per class and in-domain vs cross-domain.
4. **Explainability** layered on top of predictions.

We do not claim:

- That our weak labels equal expert labels.
- That the model can replace human investigators. It is a **decision-support / triage** tool.
- Novel model architectures. The novelty is the combination and the cross-domain evaluation.

> Saying clearly what you do *not* claim makes you look more credible in a viva.

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8. Industry Value (how this helps in real terms)

Who	How it helps
Airlines / safety departments	Automatically triage thousands of reports and surface recurring human-factor patterns (e.g., perceptual errors on approach), shortening the time from report to corrective action.
Regulators & investigators	Consistent first-pass tagging that reduces subjectivity between investigators; faster trend reports.
Training organizations	Link error types and flight phases to targeted simulator and CRM training.
Safety-management systems (SMS)	Plug the classifier in as a module to keep incident databases structured.
Other industries	The same approach (weak labels, transfer, explainability) applies to healthcare incident reports, rail, maritime, oil & gas, manufacturing safety logs.

Why the cross-domain angle matters to industry: real organizations often have lots of unlabeled voluntary-style reports and few labeled investigations. If a model trained on one source generalizes to the other, labeling cost drops. If it doesn't, we can quantify *how much* it degrades and why. Both outcomes are useful findings.

Why explainability matters: in safety-critical fields, regulators and analysts must audit decisions. SHAP/top-word outputs let them check the model is looking at meaningful words, not artifacts.

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9. Who Should Explain What (suggested split)

Assumes **4 members.** If you have 3 or 5, merge or split the rows. Rename A–D with real names. The idea: each person owns a *stage* and can say "I built/ran this part".

Member	Owns	Files	Papers they must know	Likely teacher questions
Member A — Data & Labeling Lead	Problem, datasets, weak labels	`data_loading.py`, `keyword_weak_labeler.py`, `weak_labeler_cluster.py`	Liu (2025), Madeira (2021), Macedo (2026)	What are ASRS and NTSB? How did you get labels? Why discard keywords? Why 12 of 15?
Member B — Baselines & Evaluation Lead	Preprocessing, classical ML, metrics	`preprocessing.py`, `baseline_models.py`, `evaluation.py`	Madeira (2021), Shi (2022), Vempati (2023)	Why TF-IDF? Why Micro-F1 and not accuracy? What does the sanity check show?
Member C — Deep Learning & LLM Lead	Transformers and LLM chain	`transformer_model.py`, `llm_cot_baseline.py`, `model_comparison.py`	Andrade & Walsh (2023), Liu (2025)	DistilBERT vs SafeAeroBERT? Why binary relevance? What is HFACS-CoT+?
Member D — Explainability, Analysis & Report Lead	Tuning, SHAP, phase analysis, final story	`explainability.py`, `hyperparameter_tuning.py`, `flight_phase_analysis.py`, final report	Madeira (2021), Vempati (2023), Andrade & Walsh (2023)	How does SHAP work? Which hyperparameters mattered? Why does transfer degrade? Industry value?

Rules for presentation day

- **Everyone** knows Sections 1–4 (pitch, terms, problem, pipeline). Any member can be asked these.
- Member A opens (problem + data), B covers baselines + metrics, C covers advanced models, D closes with results, explainability, industry value and limitations.
- If a question is outside your part, say: *"That part was handled by [name], but the idea is…"* and give a one-line answer. Never stay silent.
- **Cross-train:** each member should also read the *one-line summary* of the other members' files (Section 6).

[CONFIRM] Replace this split with the real contribution of each person if the work was already divided differently. Evaluators can tell when a described role doesn't match the person.

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10. Viva Question Bank (with short answer guides)

Basics

1. **What is the project about?** → Section 1.
2. **What is HFACS and why use it?** → Standard taxonomy for human-factor causes; widely used and lets us compare with literature.
3. **Difference between ASRS and NTSB?** → ASRS: voluntary, many reports, includes near-misses, self-reported bias. NTSB: official accident investigations, externally verified.
4. **Why aviation?** → High stakes, rich text, public data, well-established safety framework.
5. **What is your input and output?** → Input: report text. Output: HFACS human-factor class(es).

Data & labels

6. **Where did labels come from?** → NTSB: expert-coded HFACS in the GAHFACS-style construction. ASRS: weak labeling (keyword rule and optionally clustering). **[CONFIRM exact setup.]**
7. **What is weak labeling and is it reliable?** → Automatic noisy labels; reliability controlled by the 12/15 consistency gate; still a limitation.
8. **Why discard keywords instead of forcing a label?** → Wrong labels teach the model wrong patterns; better fewer, cleaner labels.
9. **How many reports/classes?** → **[CONFIRM numbers.]**
10. **How did you handle class imbalance?** → Used Macro-F1 along with Micro-F1 and reported per-class scores.

Methods

11. **Why TF-IDF when transformers exist?** → Fast, interpretable baseline; Shi et al. show it's competitive; tells us if complexity is justified.
12. **Why DistilBERT?** → Smaller and faster than BERT with similar performance; fits our compute. SafeAeroBERT adds aviation domain knowledge.
13. **What is binary relevance?** → One independent yes/no output per label; handles multi-label reports.
14. **What is HFACS-CoT+ and why separate calls?** → Question chain imitating an investigator; separate calls avoid logical inconsistencies, as shown by the paper's ablation.
15. **Why not just use ChatGPT?** → Cost, privacy, reproducibility; we measure how it compares to trained models instead of assuming it is better.

16. **Why UMAP + HDBSCAN?** → UMAP reduces high-dimensional embeddings; HDBSCAN needs no preset cluster count and flags outliers.

Evaluation

17. **Why Micro-F1 as primary?** → Standard in Madeira et al.; accounts for all predictions and works for multi-label.
18. **What is the cross-domain test and what did you find?** → Train on ASRS, test on NTSB. **[CONFIRM results and quote your numbers.]**
19. **Why might transfer drop?** → Reporting bias (voluntary reporters over-report perceptual errors, under-report own decision/skill failures), different writing style, different label sources.
20. **What did the sanity check compare?** → Our ASRS class distribution against Vempati's published figures, to confirm our data/labels are plausible before trusting transfer results.

Explainability & analysis

21. **How do you explain predictions?** → SHAP for transformers; top-weighted words for Logistic Regression.
22. **What did hyperparameter tuning teach you?** → Only a few hyperparameters usually matter (importance ranking); avoids wasting search effort.
23. **What does flight-phase analysis add?** → Context: errors cluster in certain phases (approach/landing), useful for training and operations.

Critical / tough ones

24. **What are the limitations?** → Weak label noise; label scheme differences between ASRS and NTSB; limited labeled data; LLM nondeterminism; English-only; reports are subjective.
25. **What is novel here?** → Not a new model; novelty is the cross-domain evaluation plus the combined, comparable pipeline with explainability.
26. **How would this be deployed?** → As a triage assistant: model suggests labels with highlighted words, human analyst confirms.
27. **Ethical concerns?** → Must not be used to blame individuals automatically; human-in-the-loop; reports may contain sensitive details.
28. **What would you do with more time?** → Expert-validate a sample of weak labels, extend to full HFACS levels (preconditions, supervision, organization), try multilingual and LLM-fine-tuning, test on other datasets.
29. **If your results were poor, what would that mean?** → Still a valid finding: it quantifies the domain gap and shows where weak labels fail.
30. **Did you copy the papers?** → No. We took specific, cited techniques (e.g., consistency criterion, HFACS-CoT+) and integrated them into a new pipeline with a cross-domain evaluation.

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11. Honest Risk Checklist (fix before the viva)

- ☐ Do all members know the final numbers (dataset sizes, best Micro-F1, cross-domain drop)?
- ☐ Can you run each script and show its output (results tables, logs, heatmap, SHAP plot)?
- ☐ Is the exact list of HFACS classes you used written down and the same everywhere? **[CONFIRM the 4 classes]**
- ☐ Did you actually complete the optional/bonus parts (`weak_labeler_cluster.py`, SafeAeroBERT, `flight_phase_analysis.py`)? Only present what you built. The plan itself says to cut Week 3 additions before Weeks 1–2 if time runs short.
- ☐ Can each person explain **one** limitation in their own words?
- ☐ Are the paper citations consistent in the report and the slides?
- ☐ Have you done one full mock viva where members ask each other Section 10 questions?

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12. Cheat Sheet (last-minute revision)

- **Goal:** classify human-factor causes in aviation reports (HFACS) and test ASRS → NTSB transfer.
- **Data:** ASRS (voluntary, abundant) and NTSB (official, expert-coded).
- **Labels:** keyword weak labels with a 12-of-15 consistency gate (+ optional clustering).
- **Models:** TF-IDF/Word2Vec/Doc2Vec + LogReg/LinearSVC → DistilBERT/SafeAeroBERT → LLM HFACS-CoT+.
- **Metrics:** Micro-F1 (primary), Macro-F1, Precision, Recall; in-domain vs cross-domain.
- **Explainability:** SHAP + top words.
- **Extras:** hyperparameter importance, flight-phase heatmap.
- **Value:** faster, consistent safety triage; lower labeling cost; auditable predictions.
- **Honesty line:** "Decision-support tool, not a replacement for investigators; weak labels are a stated limitation."